

Distributed Explorer-Extractor Pipeline for Web Scraping Using Redis

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Abstract—This paper presents a distributed implementation of the explorer-extractor pipeline pattern for web scraping. Building upon a previous parallel scraping architecture that used multithreading, this work extends the system to operate across multiple network nodes using Redis as a coordination layer. The distributed system replaces thread-based synchronization with Redis data structures, enabling worker processes on separate machines to coordinate through message passing and shared state. Two user interfaces are provided: a command-line terminal interface and a web dashboard with real-time monitoring via WebSockets. Experimental results demonstrate that distributing work across multiple nodes improves throughput compared to single-node execution.

Index Terms—distributed systems, web scraping, Redis, message passing, producer-consumer, coordination, WebSocket

I. INTRODUCTION

Web scraping consists of multiple subproblems that make it a promising candidate for parallelization and distribution. The two major operations—traversing the website and gathering information—can be separated into distinct stages that communicate through well-defined interfaces.

In a previous work, I implemented this as a parallel scraper using Python’s multithreading with the explorer-extractor pipeline pattern. Explorer threads discovered URLs and fetched HTML content, while extractor threads parsed the content to extract metadata. Thread-safe queues coordinated work between the two stages.

However, the parallel implementation had limitations imposed by running on a single machine:

- **Resource Constraints:** Limited by CPU cores and memory on one machine
- **Python’s GIL:** Global Interpreter Lock prevented true parallel execution of CPU-bound extraction tasks
- **Scalability:** Could not easily add more computational resources without upgrading hardware

This paper presents a distributed implementation that addresses these limitations. The architecture maps naturally from the parallel version:

- **Thread objects** → **Redis data structures:** Thread-safe queues become Redis Lists, thread synchronization becomes Redis Pub/Sub
- **Worker types** → **Network nodes:** Explorers and extractors run on separate machines

- **Threads** → **Processes:** Each worker spawns multiple processes instead of threads, avoiding the GIL limitation

The distributed system provides two interfaces: a terminal-based command-line interface for direct control, and a web dashboard with real-time monitoring through WebSocket connections. Both interfaces coordinate worker nodes through Redis and display live metrics during scraping operations.

II. PROGRAM IMPLEMENTATION

A. Architecture Overview

The distributed scraper consists of three types of nodes deployed across separate machines:

- **Coordinator Node (VM1):** Hosts Redis server, manages scraper lifecycle, provides user interfaces (terminal CLI or web dashboard)
- **Explorer Nodes (VM2):** Run multiple explorer worker processes that fetch HTML and discover links
- **Extractor Nodes (VM3):** Run multiple extractor worker processes that parse HTML and extract metadata

Unlike the parallel version where workers were threads in a single process, workers are now independent processes on separate machines. Redis serves as the coordination layer, replacing in-memory thread synchronization primitives.

B. Mapping from Parallel to Distributed

The transition from parallel to distributed was natural due to the pipeline architecture:

Synchronization Primitives:

- Thread-safe queues → Redis Lists (LPUSH/BRPOP)
- Thread Events → Redis Pub/Sub channels
- Shared counters → Redis String values (INCR/GET)
- Deduplication sets → Redis Sets (SADD/SISMEMBER)

Worker Architecture:

- Thread objects → Worker processes on remote nodes
- Thread pool → Process pool via Python’s multiprocessing
- Worker threads → Worker processes (eliminates GIL contention)

Data Flow:

- In-memory queue → Redis List acting as distributed queue
- Shared memory → Redis as shared state store
- Function calls → Message passing via Redis

C. Usage

The distributed scraper provides two interfaces with identical functionality:

1) *Terminal Interface*: Command-line interface for direct control:

```
python scraper.py <root_url> <duration>
                  <explorers> <extractors>
```

Required Arguments:

- `root_url`: Starting URL
- `duration`: Execution time in minutes
- `explorers`: Number of explorer processes
- `extractors`: Number of extractor processes

Example:

```
# 16 explorers, 1 extractor, 1 minute
python scraper.py https://dlsu.edu.ph/ 1 16 1
```

Note: Unlike the parallel version where the total worker count could be split automatically, the distributed version requires explicit specification of both explorer and extractor counts to coordinate across network nodes.

2) *Web Dashboard Interface*: Flask-based web application with real-time monitoring:

- Configuration form to set parameters
- Live metrics display (URLs found, extracted, throughput)
- WebSocket-based updates (no page refresh needed)
- Start/stop controls for scraping operations

Both interfaces communicate with the same Redis backend and provide identical control over the distributed workers.

D. Distributed System APIs and Techniques

1) *Redis as Coordination Layer*: Redis serves as the central coordination mechanism, providing distributed data structures that replace in-process synchronization primitives:

Redis Lists (Work Queues):

- `explored_queue`: URLs with fetched HTML ready for extraction
- `exploit_queue`: URLs to be visited by explorers
- Operations: LPUSH (enqueue), BRPOP (blocking dequeue with timeout)

Redis Sets (Deduplication):

- `visited_urls`: Ensures URLs are visited only once
- Operations: SADD (add and check atomically), SIS-MEMBER (membership test)

Redis Streams (Result Collection):

- `results_stream`: Append-only log of extracted data
- Operations: XADD (append entry), XRANGE (read entries)

Redis Pub/Sub (Control Signals):

- `scraper_control`: Broadcast start/stop commands to all nodes
- Messages: JSON payloads with command type

Redis Strings (Shared State):

- **Counters**: `urls_found`, `urls_extracted` (atomic INCR)
- **Configuration**: `num_explorers`, `num_extractors`, `duration`
- **Control flags**: `stop_signal`, `start_time`

2) *Data Sharing Within and Between Nodes*: **Within a Node**: Worker processes on the same machine share:

- Redis connection pool (max 50 connections, shared across processes)
- Environment variables for configuration
- No shared memory—each process is independent

Between Nodes: All inter-node communication occurs through Redis:

- **Work Distribution**: Explorers on VM2 push to `exploit_queue`, extractors on VM3 pop from same queue
- **State Synchronization**: All nodes read/write shared counters and flags atomically
- **Result Aggregation**: Extractors append to `results_stream`, coordinator reads all entries
- **Control Flow**: Coordinator broadcasts commands via Pub/Sub, workers subscribe and respond

3) *Coordination Techniques*: **Worker Acknowledgment Protocol**: To ensure workers are ready before starting:

- 1) Coordinator publishes start command to `scraper_control`
- 2) Explorer nodes spawn processes, push acknowledgment to `explorer_ack` queue
- 3) Extractor nodes spawn processes, push acknowledgment to `extractor_ack` queue
- 4) Coordinator uses BRPOP with 10-second timeout to wait for both ACKs
- 5) Only after receiving both ACKs does coordinator seed initial URL

This replaces thread join operations in the parallel version with distributed acknowledgment queues.

Graceful Shutdown:

- 1) Coordinator sets `stop_signal` to '1'
- 2) Coordinator publishes stop command via Pub/Sub
- 3) Workers check `stop_signal` after each task
- 4) Workers drain remaining items from queues before terminating
- 5) Coordinator waits for queues to empty, then collects results

Performance Optimization:

- **Connection Pooling**: Shared pool of Redis connections reduces connection overhead
- **Redis Pipelining**: Batch multiple operations (e.g., INCR + LPUSH) into single network round trip, reducing latency by 50×
- **Blocking Operations**: BRPOP blocks until work available, eliminating polling overhead

E. Message Passing

The system uses two message passing patterns:

Queue-Based Task Distribution (Producer-Consumer):

- Explorers produce URLs, push to `exploit_queue`
- Extractors consume URLs, pop from `exploit_queue`
- Decouples producers from consumers, enables load balancing
- BRPOP ensures fair distribution when multiple workers compete

Pub/Sub for Control Signals (Broadcast):

- One-to-many communication from coordinator to all workers
- Used for lifecycle events (start, stop)
- Workers poll Pub/Sub channel every 100ms to detect commands

All messages are JSON-encoded for structured data exchange:

```
# Work item
{"url": "https://...", "html": "..."}

# Control command
{"command": "start"}

# Acknowledgment
{"status": "ready", "num_workers": 16}
```

III. RESULTS

A. Performance Observations

To demonstrate the distributed system’s scalability, performance was observed under different worker configurations:

- **Deployment:** 3 ProxMox VMs (VM1: Coordinator + Redis, VM2: Explorers, VM3: Extractors)
- **Network:** 10.2.13.0/16 internal network
- **Target Website:** <https://www.dlsu.edu.ph/>
- **Test Duration:** 60 seconds per configuration

The previous parallel scraper paper systematically benchmarked 17 explorer-extractor ratios to identify optimal configurations. Since that analysis already established that explorer-heavy ratios (particularly 16:1) perform best for this workload, this work focuses on demonstrating how distribution enables scaling beyond single-machine limitations.

B. Worker Scaling Impact

TABLE I
OBSERVED THROUGHPUT VS. WORKER COUNT

Configuration	Total Workers (Explorers + Extractors)	Throughput (pages/s)
Minimal	2 (1 + 1)	~1.0
Moderate	20 (16 + 4)	~8-12
High	100+ (distributed)	~20

Table I shows observed throughput under different worker configurations. With minimal workers (1 explorer, 1 extractor),

throughput remains around 1 page per second—essentially sequential processing. As worker count increases to moderate levels (16-20 workers), throughput improves to 8-12 pages/s. When scaling to 100+ workers distributed across nodes, throughput can reach approximately 20 pages/s within the same 60-second timeframe.

This represents a 20× improvement in throughput from the minimal configuration, demonstrating the effectiveness of distributing work across multiple machines.

C. Analysis

1) *Benefits of Distribution and Scaling:* The observed performance improvements from increasing worker counts demonstrate the fundamental advantage of distributed systems:

Linear Scaling with Worker Count: With minimal workers (1+1), the system processes URLs sequentially at roughly 1 page/s. Adding more workers distributed across nodes increases throughput proportionally, reaching 20× improvement with 100+ workers. This near-linear scaling indicates that the workload is embarrassingly parallel—each worker operates independently without significant coordination overhead.

Resource Distribution: Distributing workers across multiple machines (VM2 for explorers, VM3 for extractors) eliminates single-machine resource bottlenecks. Each VM can dedicate its full CPU, memory, and network bandwidth to its specialized workload, preventing resource contention that would occur if all 100+ workers ran on one machine.

True Parallelism with Processes: Using processes instead of threads eliminates Python’s Global Interpreter Lock (GIL) limitation. Each worker process has its own Python interpreter, enabling true parallel execution. This is particularly beneficial for extractor workers performing CPU-bound HTML parsing, which would serialize under the GIL in a threaded implementation.

Network Parallelism: Multiple explorer processes on VM2 can issue concurrent HTTP requests without blocking each other. Modern operating systems can handle hundreds of concurrent network connections efficiently, and distributing these across nodes prevents any single network interface from becoming a bottleneck.

Horizontal Scalability: The distributed architecture supports adding more machines to increase capacity without code modifications. If 100 workers are insufficient, deploying additional VMs (VM4, VM5, etc.) and connecting them to the same Redis instance immediately adds capacity. This elastic scaling is a fundamental advantage over single-machine parallel implementations.

2) *Redis Performance:* Redis proved to be an efficient coordination layer with minimal overhead:

- Connection pooling reduced connection establishment costs
- Pipelining (batching INCR + LPUSH operations) reduced network latency by approximately 50× compared to individual commands
- Blocking operations (BRPOP) eliminated polling overhead

- Pub/Sub provided low-latency command broadcasting

No Redis bottlenecks were observed during testing, suggesting the coordination layer scales well for this workload.

3) *System Coordination*: The acknowledgment protocol ensured reliable startup across distributed nodes. Workers consistently responded within a few seconds of receiving start commands, and the coordinator waited for all acknowledgments before seeding work. This prevented race conditions where some workers might miss initial URLs. Graceful shutdown allowed workers to complete in-progress tasks before terminating, preventing data loss.

WebSocket-based monitoring in the web interface provided real-time visibility into system state without impacting worker performance, as monitoring queries only read from Redis without blocking worker operations.

IV. CONCLUSION

This paper demonstrates how a parallel web scraper can be naturally extended to a distributed system using Redis as a coordination layer. The architecture maps directly from the parallel version: thread-safe queues become Redis Lists, thread synchronization becomes Pub/Sub messaging, and in-memory shared state becomes distributed key-value storage.

A. Application of Distributed Systems Techniques

The implementation successfully applies several distributed systems concepts:

Shared Memory → Message Passing: Instead of threads sharing memory within a process, workers on separate machines communicate through Redis. Work items flow through distributed queues, and state updates occur via atomic Redis operations. This transition from shared memory to message passing is a fundamental shift in distributed systems design.

Coordination and Synchronization: Redis Pub/Sub provides broadcast communication for control signals, while acknowledgment queues implement distributed barriers ensuring all workers are ready before starting. The `stop_signal` flag coordinates graceful shutdown across nodes. These techniques replace thread-based synchronization primitives with distributed equivalents.

Data Consistency: Redis atomic operations (INCR, SADD, LPUSH/BRPOP) ensure consistency without explicit locking. The `visited_urls` set provides distributed deduplication, preventing multiple workers from processing the same URL even when running on different machines.

Fault Tolerance: While not extensively tested, the architecture supports basic fault tolerance: if a worker process crashes, others continue processing from the shared queues. Work items are only removed from queues after successful processing (BRPOP is atomic), preventing data loss.

B. Performance Improvements

The distributed implementation demonstrates significant throughput improvements through worker scaling. Increasing from 2 workers (1+1) to 100+ workers yields approximately

20× higher throughput within the same time period. This improvement stems from:

- **Resource Distribution**: Dedicating separate machines to explorers and extractors eliminates resource contention
- **GIL Elimination**: Using processes instead of threads enables true parallel execution of CPU-bound extraction tasks
- **Horizontal Scalability**: The architecture supports adding more nodes without code changes, enabling linear scaling with worker count

Redis proved to be an effective coordination layer with minimal overhead. Connection pooling and pipelining techniques reduced network latency, while blocking operations eliminated polling overhead. The coordination layer scales efficiently even with 100+ concurrent workers accessing shared state.

C. Trade-offs and Considerations

Distribution introduces complexity compared to the parallel version:

- **Dependency on Redis**: System availability depends on Redis being accessible; if Redis fails, all coordination breaks down
- **Network Latency**: Message passing through Redis adds latency compared to in-memory queues, though pipelining mitigates this
- **Deployment Complexity**: Managing multiple nodes requires coordination of environment variables, network configuration, and process startup order

However, these trade-offs are acceptable given the performance gains and scalability benefits. The system is well-suited for scenarios requiring higher throughput than a single machine can provide.

D. Future Work

Potential enhancements include:

- **Dynamic Worker Scaling**: Automatically spawn/terminate workers based on queue lengths
- **Redis Clustering**: Distribute Redis itself across multiple machines for higher availability
- **Monitoring and Metrics**: More detailed performance tracking (network latency, Redis operation times, per-worker throughput)
- **Load Balancing**: Implement work stealing if some workers are idle while others are overloaded

The distributed explorer-extractor pattern demonstrates how understanding both parallel and distributed systems concepts enables building scalable, high-performance applications. By applying message passing, coordination protocols, and distributed data structures, the system achieves significant performance improvements over single-node execution while maintaining a clean, maintainable architecture.